



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No. 1

- 1. Experiment Title:** Demonstrate Inline, Internal, External JavaScript, use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

Internal:

```
<!DOCTYPE html>

<html>

<head>

    <title>Hello World</title>

</head>

<body>

    <div>

        <h1>Hello World</h1>

        <p>Welcome to Symbiosis Institute of Technology, Nagpur</p>

    </div>

</body>

</html>
```



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```
font-size:20px;

color:#333333;

background-color:#ffffcc;

padding:10px;

margin-top:20px;

}

button{

font-size:18px;

color:white;

background-color:green;

padding:12px 24px;

border:none;

border-radius:5px;

cursor:pointer;

margin-top:20px;

}

</style>

</head>

<body>

<h1>Hello World</h1>

<p>Name : Aaryan K. Kawale , PRN : 24070521162</p>

<p>This is a sample Webpage made using External css.</p>

<button>Click Me</button>

</body>

</html>
```



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External:

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>


  <h1>Hello World</h1>

  <p> Name : Shubhansu Majare , PRN: 24070521170</p>

  <p> Sec: C, Sem: 5th </p>


  <p>This is a sample Webpage is made by using the External CSS.</p>


  <button>Click Me</button>


</body>

</html>
```

Inline:

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

</head>

<body style="background-color:#f0f8ff; text-align:center; font-family:Arial, sans-serif;
padding:40px;">
```



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```
<h1 style="font-size:48px; color:white; background-color:#007BFF; padding:15px;
border-radius:8px;">
```

Hello World

```
</h1>
```

```
<p>
```

Name : Aaryan K. Kawale ,PRN : 24070521162

```
</p>
```

```
<p style="font-size:20px; color:#333333; background-color:#ffffcc; padding:10px;
margin-top:20px;">
```

This is a sample webpage designed by Inline CSS.

```
</p>
```

```
<button style="font-size:18px; color:white; background-color:green; padding:12px
24px; border:none; border-radius:5px; cursor:pointer; margin-top:20px;">
```

Click Me

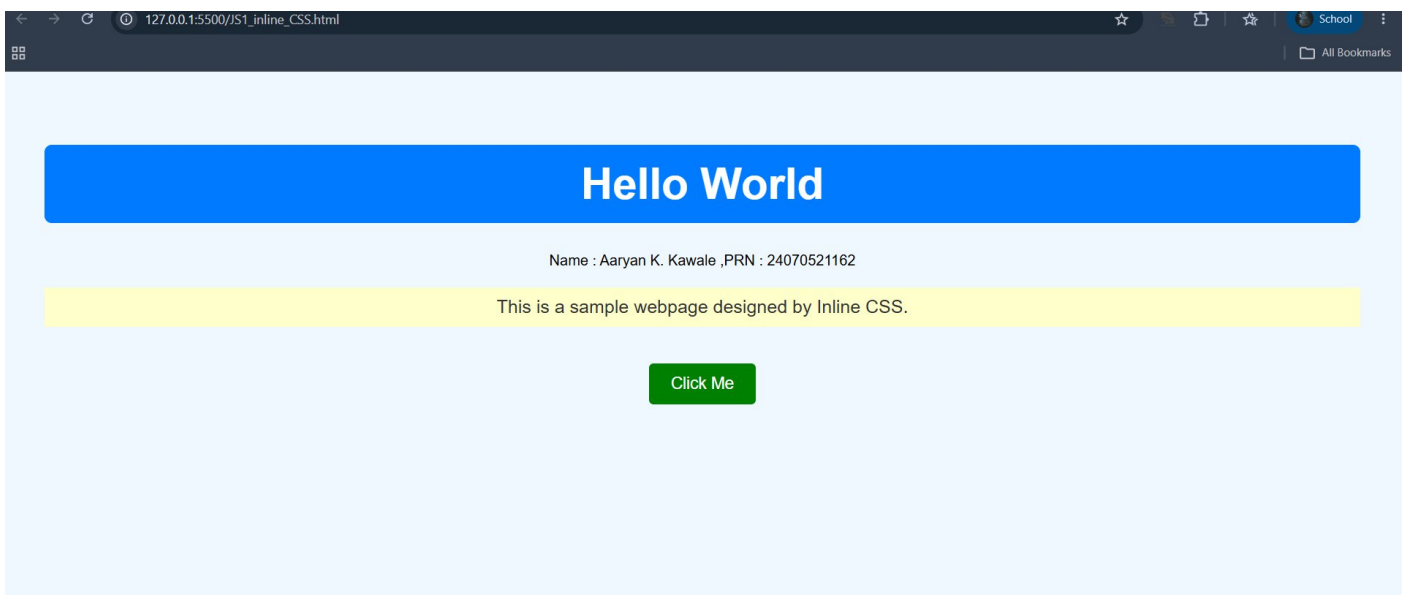
```
</button>
```

```
</body>
```

```
</html>
```

4. Output:

Inline:





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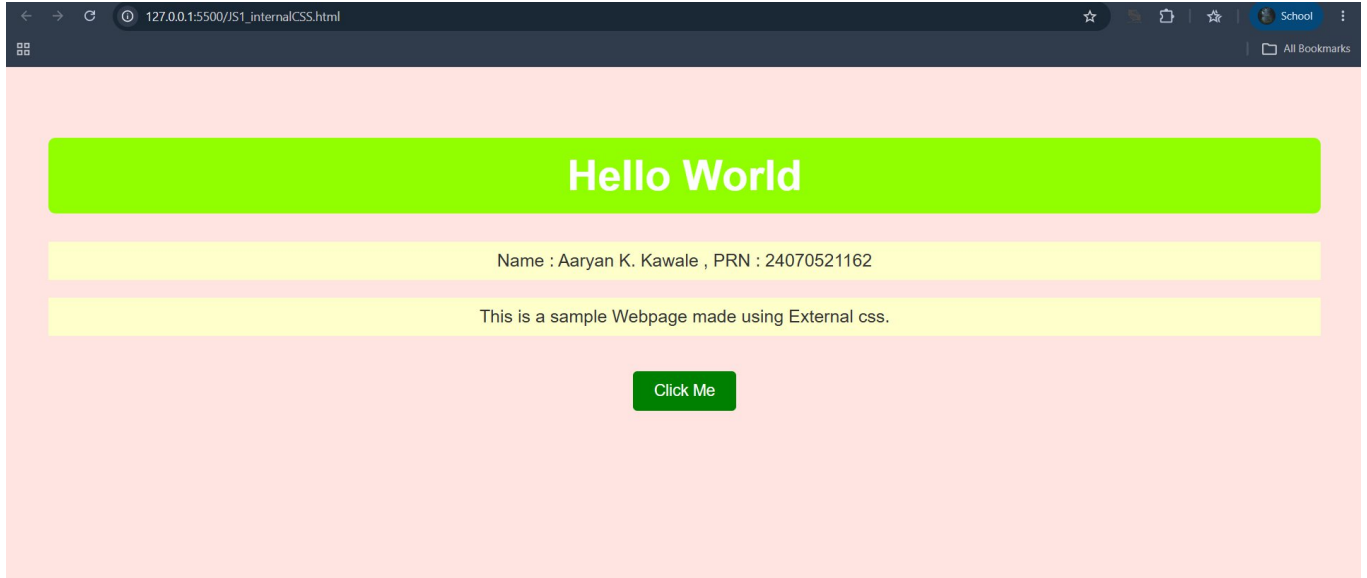
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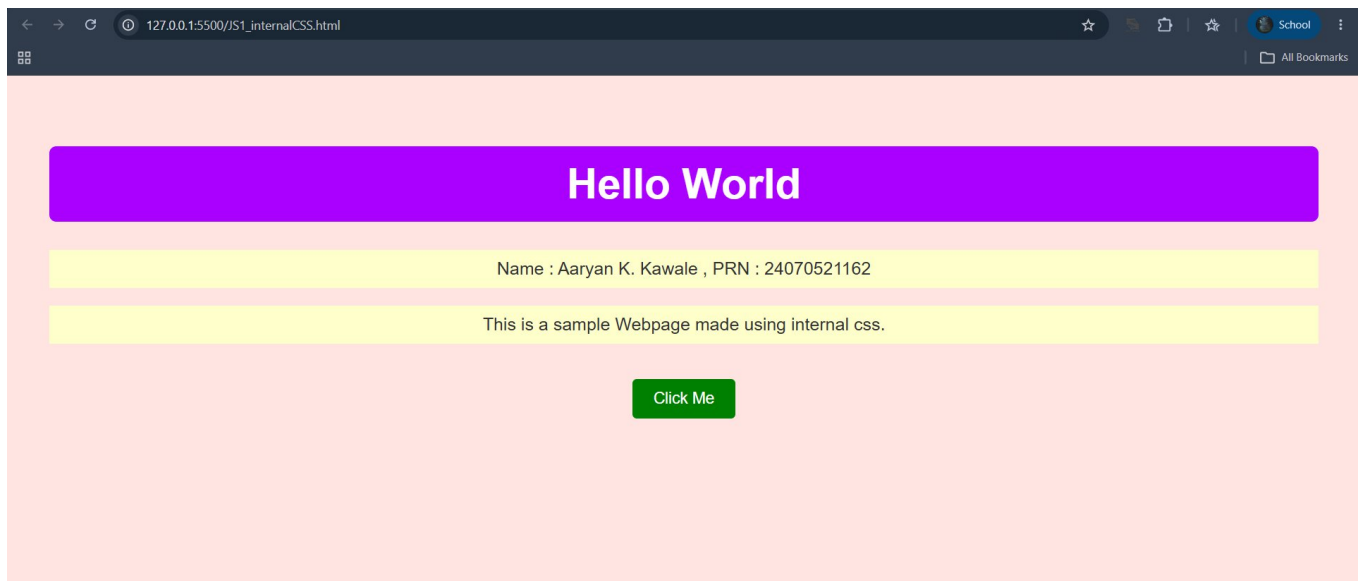
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External:



Internal:





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Case Study 1

1. **Case Study Title:** Desing a webpage for SIT Nagpur with buttons , text , images, etc...

2. **Case Study Program Code:**

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-scale=1.0">
8.   <title>SIT Nagpur - Computer Science Department</title>
9.   <style>
10.    body {
11.      font-family: Arial, sans-serif;
12.      margin: 0;
13.      padding: 0;
14.      /* Live campus background image */
15.      background-image:
16.        url('https://d3lzcn6mbbadaf.cloudfront.net/media/details/ANI-
17.        20230411114128.jpg');
18.      background-size: cover;
19.      background-position: center;
20.      background-attachment: fixed;
21.      background-repeat: no-repeat;
22.    }
23.    header {
24.      background-color: #ae1717b8; /* Symbiosis Maroon */
25.      color: white;
26.      padding: 20px;
27.      text-align: center;
28.    }
29.    .logo {
30.      max-width: 180px;
31.      height: auto;
32.      margin-bottom: 10px;
33.      background-color: transparent;
34.    }
35.    nav {
36.      background-color: #333;
37.      overflow: hidden;
38.    }
39.    nav a {
40.      float: left;
41.      display: block;
42.      color: white;
43.      text-align: center;
44.      padding: 14px 20px;
```



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```
43.     text-decoration: none;
44. }
45. nav a:hover {
46.     background-color: #ddd;
47.     color: black;
48. }
49.
50. /* Flexbox wrapper to place sections side by side */
51. .content-wrapper {
52.     display: flex;
53.     flex-direction: row;
54.     justify-content: center;
55.     align-items: stretch;
56.     gap: 30px;
57.     max-width: 1200px;
58.     margin: 50px auto 100px auto;
59.     padding: 0 20px;
60. }
61.
62. /* Shared stylings for both white info blocks */
63. .info-box {
64.     flex: 1;
65.     padding: 30px;
66.     background: rgba(255, 255, 255, 0.92);
67.     border-radius: 8px;
68.     box-shadow: 0 4px 15px rgba(0,0,0,0.3);
69. }
70.
71. h2, h3 {
72.     color: #333;
73. }
74. ul {
75.     padding-left: 20px;
76.     line-height: 1.6;
77. }
78. li {
79.     margin-bottom: 8px;
80. }
81.
82. footer {
83.     text-align: center;
84.     padding: 15px;
85.     background-color: #333;
86.     color: white;
87.     position: fixed;
88.     bottom: 0;
89.     width: 100%;
90. }
```



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```
91.
92.  /* Responsive design for tablets and mobile viewports */
93.  @media (max-width: 768px) {
94.      .content-wrapper {
95.          flex-direction: column;
96.          margin-top: 20px;
97.      }
98.  }
99. </style>
100. </head>
101. <body>
102.
103.     <header>
104.         
107.         <h1>Department of Computer Science & Engineering</h1>
108.         <p>Symbiosis Institute of Technology, Nagpur</p>
109.     </header>
110.
111.     <nav>
112.         <a href="https://sitnagpur.edu.in/">Home</a>
113.         <a href="student_info.html">Student Information</a>
114.     </nav>
115.
116.     <div class="content-wrapper">
117.
118.         <div class="info-box">
119.             <h2>Welcome to the Department</h2>
120.             <p>Welcome to the Department of CSE at SIT Nagpur. We are
121.             committed to delivering world-class engineering education, fostering innovation,
122.             and preparing our students to lead in the global technology landscape.</p>
123.
124.             <h3>Key Academic Highlights</h3>
125.             <ul>
126.                 <li>State-of-the-art Research & Computing Labs</li>
127.                 <li>Industry-aligned & Integrated Curriculum</li>
128.                 <li>Excellent Placement Support & Recruitment Cells</li>
129.             </ul>
130.         </div>
131.
132.         <div class="info-box">
133.             <h2>Campus Infrastructure</h2>
134.             <p>SIT Nagpur provides premium, modern amenities designed to
135.             enhance student living, academic research, and holistic wellness across the
136.             campus footprint.</p>
137.
138.             <h3>Facilities Provided</h3>
```




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```
133.         <ul>
134.             <li><strong>Academic Blocks:</strong> Smart, fully air-
                conditioned digital classrooms.</li>
135.             <li><strong>Central Library:</strong> Digital library ecosystem
                packed with global e-resources.</li>
136.             <li><strong>Living Amenities:</strong> Wi-Fi enabled secure
                hostels for girls & boys.</li>
137.             <li><strong>Dining Options:</strong> In-house food mess
                alongside the modern "Symbi-Eats" cafeteria.</li>
138.             <li><strong>Sports & Gym:</strong> High-end fitness setups,
                sports complex, and wide athletic grounds.</li>
139.         </ul>
140.     </div>
141.
142. </div>
143.
144. <footer>
145.     &copy; 2026 Symbiosis Institute of Technology, Nagpur. All Rights
        Reserved.
146. </footer>
147.
148. <script src="script.js"></script>
149. </body>
150. </html>
```



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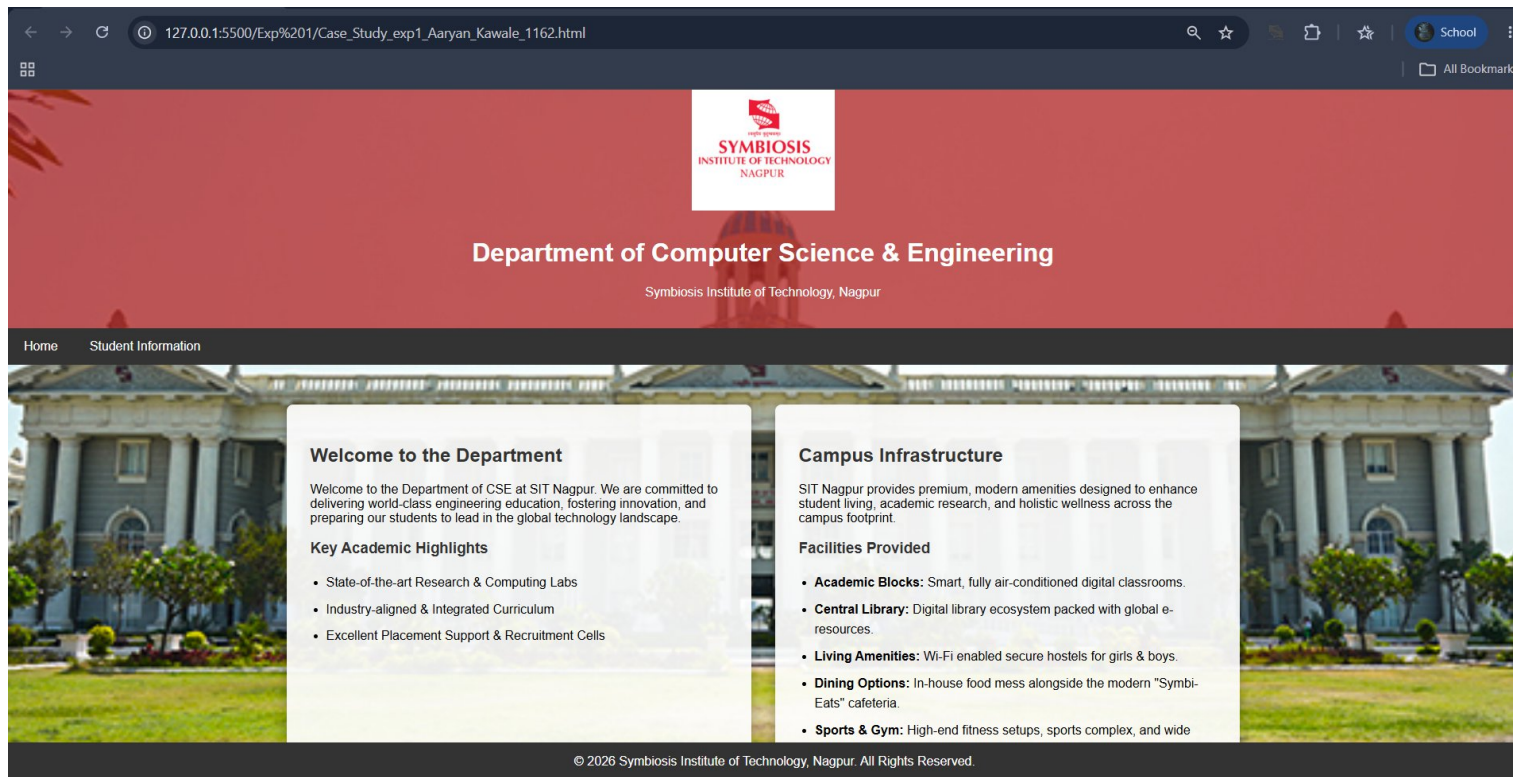
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3.Output:



Experiment No. 2



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- 1. Experiment Title:** Use var,let,const and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html>
6. <head>
7.   <title>Bill Receipt</title>
8.   <style>
9.     body {
10.      background-color: #f4f4f9;
11.      font-family: Arial, sans-serif;
12.      display: flex;
13.      justify-content: center;
14.      align-items: center;
15.      height: 100vh;
16.      margin: 0;
17.    }
18.    .receipt-box {
19.      background-color: #ffffff;
20.      border: 2px solid #333333;
21.      padding: 30px;
22.      width: 350px;
23.      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
24.    }
25.    h3 {
26.      text-align: center;
27.      margin-top: 0;
28.    }
29.  </style>
30. </head>
31. <body>
32.
33.  <script>
34.    // 1. Popups to take user input
35.    var uName = prompt("Enter User Name:");
36.    var contact = prompt("Enter Contact No:");
37.    var pName = prompt("Enter Product Name:");
38.    var qty = Number(prompt("Enter Quantity:"));
39.    var price = Number(prompt("Enter Price per Unit:"));
40.
41.    // 2. Mathematical calculations
```



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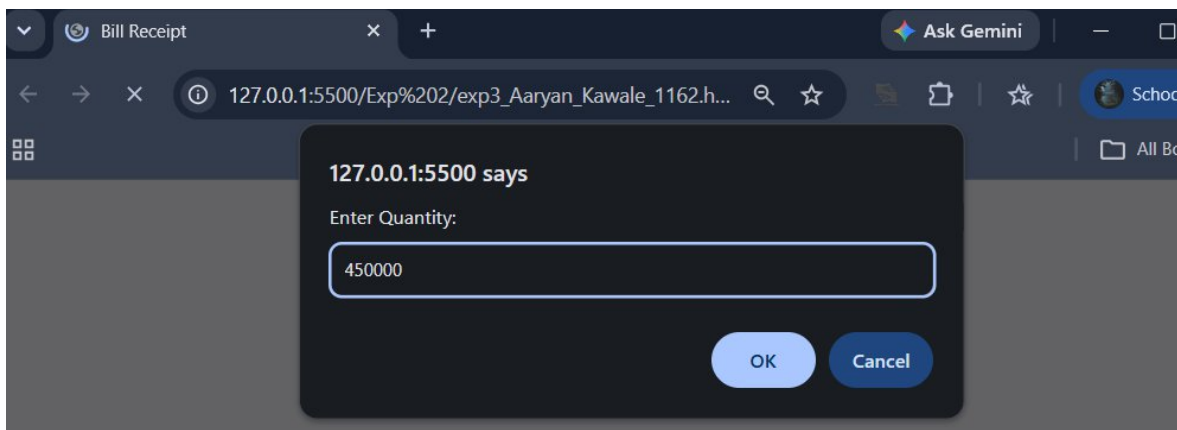
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```
42. var total = qty * price;
43. var gst = total * 0.18;
44. var finalAmount = total + gst;
45.
46. // 3. Outputting only the centered bill box
47. document.write("<div class='receipt-box'>");
48. document.write("<h3>Bill Receipt</h3>");
49. document.write("<p><b>User Name:</b> " + uName + "</p>");
50. document.write("<p><b>Contact No:</b> " + contact + "</p>");
51. document.write("<p><b>Product Name:</b> " + pName + "</p>");
52. document.write("<p><b>Quantity:</b> " + qty + "</p>");
53. document.write("<p><b>Price:</b> " + price + "</p>");
54. document.write("<hr>");
55. document.write("<p><b>Total Value:</b> " + total + "</p>");
56. document.write("<p><b>GST (18%):</b> " + gst + "</p>");
57. document.write("<p style='font-size: 18px;'><b>Final Amount:</b> " +
  finalAmount + "</p>");
58. document.write("<br>");
59. document.write("<p style='font-size: 12px; color: #555; text-align:
  center;'>Name: Aaryan Kawale, PRN: 24070521162</p>");
60. document.write("</div>");
61. </script>
62.
63. </body>
64. </html>
```

4.Output:





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127.0.0.1:5500 says

Enter Contact No:

24070521162

OK Cancel

127.0.0.1:5500 says

Enter User Name:

Aaryan Kawale

OK Cancel

Bill Receipt

User Name: Aaryan Kawale

Contact No: 24070521162

Product Name: Asus

Quantity: 5

Price: 40000

Total Value: 200000

GST (18%): 36000

Final Amount: 236000

Name: Aaryan Kawale, PRN: 24070521162



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Case Study 2

1. **Case Study Title:** Develop the webpage for an online shopping website using html, css js .

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6. <meta charset="UTF-8">
7. <meta name="viewport" content="width=device-width, initial-scale=1.0">
8. <title>Simple Shop</title>
9. <style>
10. * { box-sizing: border-box; margin: 0; padding: 0; font-family: sans-serif; }
11.
12. /* Split layout: Left column for items, right column for cart */
13. body { display: flex; min-height: 100vh; background: #f4f4f9; }
14.
15. .products-section { flex: 2; padding: 20px; }
16. .cart-section { flex: 1; padding: 20px; background: #ffffff; border-left: 2px solid
  #ddd; }
17.
18. h2 { margin-bottom: 15px; color: #333; }
19.
20. /* Product Grid */
21. .grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(180px, 1fr));
  gap: 15px; }
22. .card { background: #fff; padding: 15px; border-radius: 8px; border: 1px solid
  #ddd; text-align: center; }
23. .card h3 { font-size: 1.1rem; margin-bottom: 5px; }
24. .price { color: #2563eb; font-weight: bold; margin-bottom: 10px; }
25.
26. button { background: #2563eb; color: white; border: none; padding: 8px 12px;
  border-radius: 5px; cursor: pointer; }
27. button:hover { background: #1d4ed8; }
28.
29. /* Cart Table & Summary */
30. table { width: 100%; border-collapse: collapse; margin-bottom: 20px; }
31. th, td { text-align: left; padding: 8px; border-bottom: 1px solid #eee; }
32. .remove-btn { background: #ef4444; padding: 4px 8px; font-size: 0.8rem; }
33. .remove-btn:hover { background: #dc2626; }
34.
35. .total-box { font-size: 1.2rem; font-weight: bold; display: flex; justify-content:
  space-between; margin-top: 15px; }
```



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```
36. .checkout-btn { width: 100%; padding: 10px; background: #10b981; margin-top:
    15px; font-size: 1rem; }
37. .checkout-btn:hover { background: #059669; }
38. </style>
39. </head>
40. <body>
41.
42. <!-- LEFT COLUMN: ITEMS -->
43. <section class="products-section">
44.   <h2>Available Items</h2>
45.   <div class="grid" id="productGrid"></div>
46. </section>
47.
48. <!-- RIGHT COLUMN: CART & TOTAL -->
49. <section class="cart-section">
50.   <h2>Shopping Cart</h2>
51.   <table>
52.     <thead>
53.       <tr>
54.         <th>Item</th>
55.         <th>Qty</th>
56.         <th>Price</th>
57.         <th></th>
58.       </tr>
59.     </thead>
60.     <tbody id="cartTable"></tbody>
61.   </table>
62.
63.   <div class="total-box">
64.     <span>Total:</span>
65.     <span id="grandTotal">₹0.00</span>
66.   </div>
67.
68.   <button class="checkout-btn" onclick="checkout()">Checkout</button>
69. </section>
70.
71. <script>
72.   // 1. Items Data (Prices in INR)
73.   const products = [
74.     { id: 1, name: "Wireless Mouse", price: 1428.00 },
75.     { id: 2, name: "Keyboard", price: 2856.00 },
76.     { id: 3, name: "Headphones", price: 4284.00 },
77.     { id: 4, name: "USB Cable", price: 761.60 }
78.   ];
79.
80.   let cart = [];
81.
82.   // 2. Render Product Cards (Left Column)
```



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```
83. function renderProducts() {
84.   const grid = document.getElementById("productGrid");
85.   grid.innerHTML = products.map(p => `
86.     <div class="card">
87.       <h3>${p.name}</h3>
88.       <p class="price">₹${p.price.toFixed(2)}</p>
89.       <button onclick="addToCart(${p.id})">Add to Cart</button>
90.     </div>
91.   `).join("");
92. }
93.
94. // 3. Add to Cart Logic
95. function addToCart(productId) {
96.   const item = cart.find(i => i.id === productId);
97.   if (item) {
98.     item.qty += 1;
99.   } else {
100.     const product = products.find(p => p.id === productId);
101.     cart.push({ ...product, qty: 1 });
102.   }
103.   renderCart();
104. }
105.
106. // 4. Remove from Cart Logic
107. function removeFromCart(productId) {
108.   cart = cart.filter(i => i.id !== productId);
109.   renderCart();
110. }
111.
112. // 5. Render Cart Items & Total (Right Column)
113. function renderCart() {
114.   const table = document.getElementById("cartTable");
115.   const totalEl = document.getElementById("grandTotal");
116.
117.   if (cart.length === 0) {
118.     table.innerHTML = `<tr><td colspan="4" style="text-align:center;
119.     color:#888;">Cart is empty</td></tr>`;
120.     totalEl.textContent = "₹0.00";
121.     return;
122.   }
123.   table.innerHTML = cart.map(i => `
124.     <tr>
125.       <td>${i.name}</td>
126.       <td>${i.qty}</td>
127.       <td>₹${(i.price * i.qty).toFixed(2)}</td>
128.       <td><button class="remove-btn"
129.         onclick="removeFromCart(${i.id})">X</button></td>
130.     </tr>`
131.   ).join("");
132.   totalEl.textContent = `₹${cart.reduce((total, item) => total + item.price * item.qty, 0).toFixed(2)}`;
133. }
```




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```
129.     </tr>
130.     `).join(");
131.
132.     const total = cart.reduce((sum, item) => sum + (item.price * item.qty), 0);
133.     totalEl.textContent = `₹${total.toFixed(2)} `;
134. }
135.
136. function checkout() {
137.     if (cart.length === 0) return alert("Your cart is empty!");
138.     alert("Order placed successfully!");
139.     cart = [];
140.     renderCart();
141. }
142.
143. // Initial load
144. renderProducts();
145. renderCart();
146. </script>
147.
148. </body>
149. </html>
```

3.Output:

The screenshot shows a web application interface. On the left, under 'Available Items', there are four product cards: 'Wireless Mouse' (₹1428.00), 'Keyboard' (₹2856.00), 'Headphones' (₹4284.00), and 'USB Cable' (₹761.60). Each card has an 'Add to Cart' button. On the right, under 'Shopping Cart', there is a table with columns 'Item', 'Qty', and 'Price'. The table lists the same four items with a quantity of 1 and their respective prices. Each item has a red 'X' button next to it. Below the table, the 'Total' is ₹9329.60, and there is a green 'Checkout' button.

Item	Qty	Price
USB Cable	1	₹761.60
Headphones	1	₹4284.00
Keyboard	1	₹2856.00
Wireless Mouse	1	₹1428.00

Total: ₹9329.60

Checkout

Experiment No. 3

- Experiment Title:** Implement control structures and form validations; create a grading system based on user-entered marks.



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2. Software/Tools Required: VS Code, Html/JS Compiler...

3. Experiment Program Code:

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-scale=1.0">
9.   <title>5-Subject Grading System</title>
10.  <style>
11.    * { box-sizing: border-box; font-family: system-ui, -apple-system, sans-serif; }
12.    body { background-color: #f4f6f9; display: flex; justify-content: center; align-items: center; min-height: 100vh; margin: 0; padding: 1.5rem; }
13.    .card { background: #ffffff; padding: 2rem; border-radius: 12px; box-shadow: 0 4px 16px rgba(0,0,0,0.08); width: 100%; max-width: 480px; }
14.    h2 { margin-top: 0; color: #1e293b; text-align: center; }
15.    .grid { display: grid; grid-template-columns: 1fr 1fr; gap: 0.75rem 1rem; }
16.    .form-group { margin-bottom: 0.75rem; }
17.    .full-width { grid-column: span 2; }
18.    label { display: block; margin-bottom: 0.35rem; font-weight: 600; color: #334155; font-size: 0.9rem; }
19.    input[type="number"] { width: 100%; padding: 0.65rem; border: 1px solid #cbd5e1; border-radius: 6px; font-size: 0.95rem; }
20.    input[type="number"]:focus { outline: none; border-color: #2563eb; box-shadow: 0 0 0 3px rgba(37,99,235,0.15); }
21.    button { width: 100%; padding: 0.75rem; background-color: #2563eb; color: white; border: none; border-radius: 6px; font-size: 1rem; font-weight: 600; cursor: pointer; margin-top: 0.5rem; transition: background 0.2s; }
22.    button:hover { background-color: #1d4ed8; }
23.    .error-banner { background-color: #fef2f2; border-left: 4px solid #ef4444; color: #991b1b; padding: 0.75rem; border-radius: 4px; font-size: 0.875rem; margin-bottom: 1rem; display: none; }
24.    .result-box { margin-top: 1.25rem; padding: 1.25rem; border-radius: 8px; display: none; }
25.    .result-header { font-size: 1.1rem; font-weight: 700; margin-bottom: 0.5rem; text-align: center; }
26.    .result-details { font-size: 0.95rem; line-height: 1.5; }
27.    /* Dynamic Theme Colors */
28.    .grade-A { background-color: #dcfce7; color: #166534; border: 1px solid #bbf7d0; }
29.    .grade-B { background-color: #e0f2fe; color: #075985; border: 1px solid #bae6fd; }
30.    .grade-C { background-color: #fef9c3; color: #854d0e; border: 1px solid #fef08a; }
31.    .grade-D { background-color: #ffedd5; color: #9a3412; border: 1px solid #fed7aa; }
32.    .grade-F { background-color: #fee2e2; color: #991b1b; border: 1px solid #fca5a5; }
```



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```
33. </style>
34. </head>
35. <body>
36.
37. <div class="card">
38.   <h2>Student Assessment</h2>
39.
40.   <!-- Unified Error Banner -->
41.   <div id="errorBanner" class="error-banner"></div>
42.
43.   <form id="gradeForm" novalidate>
44.     <div class="grid">
45.       <div class="form-group">
46.         <label for="sub1">Mathematics</label>
47.         <input type="number" id="sub1" class="subject-input" placeholder="0 -
100">
48.       </div>
49.       <div class="form-group">
50.         <label for="sub2">Science</label>
51.         <input type="number" id="sub2" class="subject-input" placeholder="0 -
100">
52.       </div>
53.       <div class="form-group">
54.         <label for="sub3">English</label>
55.         <input type="number" id="sub3" class="subject-input" placeholder="0 -
100">
56.       </div>
57.       <div class="form-group">
58.         <label for="sub4">History</label>
59.         <input type="number" id="sub4" class="subject-input" placeholder="0 -
100">
60.       </div>
61.       <div class="form-group full-width">
62.         <label for="sub5">Computer Science</label>
63.         <input type="number" id="sub5" class="subject-input" placeholder="0 -
100">
64.       </div>
65.     </div>
66.     <button type="submit">Calculate Final Results</button>
67.   </form>
68.
69.   <div id="resultBox" class="result-box">
70.     <div id="resultHeader" class="result-header"></div>
71.     <div id="resultDetails" class="result-details"></div>
72.   </div>
73. </div>
74.
75. <script>
```



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```
76. document.getElementById('gradeForm').addEventListener('submit', function(e) {
77.     e.preventDefault();
78.
79.     const inputs = document.querySelectorAll('.subject-input');
80.     const errorBanner = document.getElementById('errorBanner');
81.     const resultBox = document.getElementById('resultBox');
82.     const resultHeader = document.getElementById('resultHeader');
83.     const resultDetails = document.getElementById('resultDetails');
84.
85.     // Reset UI state
86.     errorBanner.style.display = 'none';
87.     resultBox.style.display = 'none';
88.     resultBox.className = 'result-box';
89.
90.     const scores = [];
91.     let hasFailedSubject = false;
92.
93.     // --- 1. CONTROL STRUCTURE: Loop with Validation Checks ---
94.     for (let i = 0; i < inputs.length; i++) {
95.         const value = inputs[i].value.trim();
96.         const label = inputs[i].previousElementSibling.textContent;
97.
98.         // Check empty input
99.         if (value === "") {
100.             showError(`Please fill in marks for <strong>${label}</strong>. `);
101.             inputs[i].focus();
102.             return;
103.         }
104.
105.         const score = Number(value);
106.
107.         // Check invalid numbers or out-of-bounds inputs
108.         if (isNaN(score) || score < 0 || score > 100) {
109.             showError(`Invalid mark for <strong>${label}</strong>. Enter a value
between 0 and 100. `);
110.             inputs[i].focus();
111.             return;
112.         }
113.
114.         // Check for subject failure condition (e.g., passing mark is 40)
115.         if (score < 40) {
116.             hasFailedSubject = true;
117.         }
118.
119.         scores.push(score);
120.     }
121.
122.     // --- 2. CALCULATIONS ---
```



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```
123.     const totalMarks = scores.reduce((sum, current) => sum + current, 0);
124.     const percentage = (totalMarks / (scores.length * 100)) * 100;
125.
126.     // --- 3. CONTROL STRUCTURE: Grading Logic ---
127.     let grade = "";
128.     let status = "";
129.     let cssClass = "";
130.
131.     if (hasFailedSubject) {
132.         // Essential academic condition: Failing any individual subject overrides
average grade
133.         status = 'FAILED (Failed 1 or more subjects)';
134.         grade = 'F';
135.         cssClass = 'grade-F';
136.     } else if (percentage >= 90) {
137.         status = 'PASSED';
138.         grade = 'A+ (Outstanding)';
139.         cssClass = 'grade-A';
140.     } else if (percentage >= 80) {
141.         status = 'PASSED';
142.         grade = 'A (Excellent)';
143.         cssClass = 'grade-A';
144.     } else if (percentage >= 70) {
145.         status = 'PASSED';
146.         grade = 'B (Good)';
147.         cssClass = 'grade-B';
148.     } else if (percentage >= 60) {
149.         status = 'PASSED';
150.         grade = 'C (Satisfactory)';
151.         cssClass = 'grade-C';
152.     } else if (percentage >= 50) {
153.         status = 'PASSED';
154.         grade = 'D (Pass)';
155.         cssClass = 'grade-D';
156.     } else {
157.         status = 'FAILED';
158.         grade = 'F (Needs Improvement)';
159.         cssClass = 'grade-F';
160.     }
161.
162.     // --- 4. RENDER OUTPUT ---
163.     resultHeader.textContent = `Status: ${status}`;
164.     resultDetails.innerHTML = `
165.         <strong>Total Marks:</strong> ${totalMarks} / 500<br>
166.         <strong>Overall Percentage:</strong> ${percentage.toFixed(2)}%<br>
167.         <strong>Final Grade:</strong> ${grade}<br>
168.         <hr style="margin: 0.75rem 0; border: none; border-top: 1px dashed
rgba(0,0,0,0.2);">
```



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```
169.         <strong>Student Name:</strong> Aaryan K. Kawale<br>
170.         <strong>PRN:</strong> 24070521162
171.         `;
172.
173.         resultBox.classList.add(cssClass);
174.         resultBox.style.display = 'block';
175.     });
176.
177.     function showError(htmlMessage) {
178.         const errorBanner = document.getElementById('errorBanner');
179.         errorBanner.innerHTML = htmlMessage;
180.         errorBanner.style.display = 'block';
181.     }
182. </script>
183. </body>
184. </html>
```

4.Output:



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Student Assessment

Mathematics	Science
85	84
English	History
85	74
Computer Science	
98	

Calculate Final Results

Status: PASSED

Total Marks: 426 / 500
Overall Percentage: 85.20%
Final Grade: A (Excellent)

Student Name: Aaryan K. Kawale
PRN: 24070521162

Case Study 3

- Case Study Title:** Desing a webpage that match password with the user password, if matched it displays



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result, if not result not displayed.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-scale=1.0">
8.   <title>Result Portal</title>
9.   <style>
10.    * {
11.      box-sizing: border-box;
12.      margin: 0;
13.      padding: 0;
14.      font-family: 'Segoe UI', sans-serif;
15.    }
16.
17.    body {
18.      display: flex;
19.      min-height: 100vh;
20.      background: #f8fafc;
21.    }
22.
23.    /* Split Screen Layout */
24.    .section-left {
25.      flex: 1;
26.      padding: 40px 20px;
27.      background: #ffffff;
28.      border-right: 2px solid #e2e8f0;
29.      display: flex;
30.      justify-content: center;
31.      align-items: center;
32.    }
33.
34.    .section-right {
35.      flex: 1;
36.      padding: 40px 20px;
37.      display: flex;
38.      justify-content: center;
39.      align-items: center;
40.    }
41.
42.    /* Form Design */
43.    .box {
44.      width: 100%;
45.      max-width: 320px;
46.    }
47.
```



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```
48. h2 {
49.   margin-bottom: 20px;
50.   color: #0f172a;
51. }
52.
53. label {
54.   display: block;
55.   font-weight: 600;
56.   margin-bottom: 8px;
57.   color: #334155;
58. }
59.
60. input {
61.   width: 100%;
62.   padding: 10px 12px;
63.   border: 1px solid #cbd5e1;
64.   border-radius: 6px;
65.   font-size: 16px;
66.   margin-bottom: 15px;
67.   outline: none;
68. }
69.
70. input:focus {
71.   border-color: #2563eb;
72. }
73.
74. button {
75.   width: 100%;
76.   padding: 10px;
77.   background: #2563eb;
78.   color: white;
79.   border: none;
80.   border-radius: 6px;
81.   font-size: 16px;
82.   font-weight: 600;
83.   cursor: pointer;
84. }
85.
86. button:hover {
87.   background: #1d4ed8;
88. }
89.
90. .error {
91.   color: #dc2626;
92.   background: #fef2f2;
93.   border: 1px solid #fecaca;
94.   padding: 8px;
95.   border-radius: 6px;
```



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```
96.   margin-top: 12px;
97.   display: none;
98.   text-align: center;
99.   font-size: 14px;
100.  }
101.
102.  /* Result Card Design */
103.  .placeholder {
104.    color: #64748b;
105.    font-size: 16px;
106.  }
107.
108.  .result-card {
109.    display: none;
110.    background: white;
111.    border: 2px solid #10b981;
112.    border-radius: 12px;
113.    padding: 30px;
114.    text-align: center;
115.    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
116.    width: 260px;
117.  }
118.
119.  .status {
120.    background: #d1fae5;
121.    color: #065f46;
122.    padding: 4px 12px;
123.    border-radius: 12px;
124.    font-weight: bold;
125.    font-size: 14px;
126.    display: inline-block;
127.    margin-bottom: 15px;
128.  }
129.
130.  .gpa {
131.    font-size: 48px;
132.    font-weight: 800;
133.    color: #2563eb;
134.  }
135.
136.  .label {
137.    color: #64748b;
138.    font-size: 14px;
139.    font-weight: 600;
140.  }
141.  </style>
142. </head>
143. <body>
```



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```
144.
145.     <!-- LEFT COLUMN: PASSWORD INPUT -->
146.     <div class="section-left">
147.         <div class="box">
148.             <h2>Student Login</h2>
149.             <form id="loginForm">
150.                 <label for="passInput">Password</label>
151.                 <!-- type="text" की वजह से पासवर्ड टाइप करते वक़्त दिखता रहेगा -->
152.                 <input type="text" id="passInput" placeholder="Enter password..."
required autocomplete="off">
153.                 <button type="submit">Submit</button>
154.             </form>
155.             <div id="errorMsg" class="error">Incorrect Password!</div>
156.         </div>
157.     </div>
158.
159.     <!-- RIGHT COLUMN: RESULT DISPLAY -->
160.     <div class="section-right">
161.         <div id="placeholder" class="placeholder">🔒 Enter correct password to
view result</div>
162.
163.         <div id="resultCard" class="result-card">
164.             <span class="status">Pass</span>
165.             <div class="gpa">9</div>
166.             <div class="label">GPA</div>
167.         </div>
168.     </div>
169.
170.     <!-- JAVASCRIPT -->
171.     <script>
172.         const TARGET_PASS = "Aaryan1162";
173.
174.         document.getElementById("loginForm").addEventListener("submit",
function(e) {
175.             e.preventDefault();
176.             const inputVal = document.getElementById("passInput").value.trim();
177.
178.             const isCorrect = (inputVal === TARGET_PASS);
179.
180.             document.getElementById("errorMsg").style.display = isCorrect ?
"none" : "block";
181.             document.getElementById("resultCard").style.display = isCorrect ?
"block" : "none";
182.             document.getElementById("placeholder").style.display = isCorrect ?
"none" : "block";
183.         });
184.     </script>
185.
```



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```
186. </body>
187. </html>
```

3.Output:

Student Login

Password

Aaryan1162

Submit

Pass

9

GPA

Student Login

Password

Aaryan24070521162

Submit

Incorrect Password!

Enter correct password to view result

Experiment No. 4

- 1. Experiment Title:** Use Function types, scope, and closures; apply try and catch; build a palindrome checker.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**



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```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-scale=1.0">
9.   <title>Palindrome Checker</title>
10.  <style>
11.    :root {
12.      --bg-color: #f4f6f8;
13.      --card-bg: #ffffff;
14.      --primary: #4f46e5;
15.      --primary-hover: #4338ca;
16.      --text-main: #1f2937;
17.      --text-muted: #6b7280;
18.      --success-bg: #d1fae5;
19.      --success-text: #065f46;
20.      --error-bg: #fee2e2;
21.      --error-text: #991b1b;
22.    }
23.
24.    * {
25.      box-sizing: border-box;
26.      margin: 0;
27.      padding: 0;
28.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica,
29.        Arial, sans-serif;
30.    }
31.    body {
32.      background-color: var(--bg-color);
33.      color: var(--text-main);
34.      display: flex;
35.      justify-content: center;
36.      align-items: center;
37.      min-height: 100vh;
38.      padding: 20px;
39.    }
40.
41.    .container {
42.      background-color: var(--card-bg);
43.      padding: 32px;
44.      border-radius: 12px;
45.      box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.1), 0 4px 6px -2px rgba(0, 0, 0,
46.        0.05);
47.      width: 100%;
48.      max-width: 480px;
49.    }
```




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```
50. h1 {
51.   font-size: 1.5rem;
52.   margin-bottom: 8px;
53.   color: var(--text-main);
54.   text-align: center;
55. }
56.
57. p.subtitle {
58.   font-size: 0.9rem;
59.   color: var(--text-muted);
60.   text-align: center;
61.   margin-bottom: 24px;
62. }
63.
64. .input-group {
65.   display: flex;
66.   flex-direction: column;
67.   gap: 12px;
68. }
69.
70. input[type="text"] {
71.   width: 100%;
72.   padding: 12px 16px;
73.   font-size: 1rem;
74.   border: 2px solid #e5e7eb;
75.   border-radius: 8px;
76.   outline: none;
77.   transition: border-color 0.2s;
78. }
79.
80. input[type="text"]:focus {
81.   border-color: var(--primary);
82. }
83.
84. button {
85.   width: 100%;
86.   padding: 12px;
87.   font-size: 1rem;
88.   font-weight: 600;
89.   color: #ffffff;
90.   background-color: var(--primary);
91.   border: none;
92.   border-radius: 8px;
93.   cursor: pointer;
94.   transition: background-color 0.2s;
95. }
96.
97. button:hover {
```



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```
98.     background-color: var(--primary-hover);
99. }
100.
101.     #result {
102.         margin-top: 20px;
103.         padding: 12px 16px;
104.         border-radius: 8px;
105.         font-size: 0.95rem;
106.         text-align: center;
107.         display: none;
108.         word-break: break-word;
109.     }
110.
111.     #result.success {
112.         display: block;
113.         background-color: var(--success-bg);
114.         color: var(--success-text);
115.     }
116.
117.     #result.error {
118.         display: block;
119.         background-color: var(--error-bg);
120.         color: var(--error-text);
121.     }
122. </style>
123. </head>
124. <body>
125.
126.     <div class="container">
127.         <h1>Palindrome Checker</h1>
128.         <p class="subtitle">Enter a word, number, or phrase to test</p>
129.
130.         <div class="input-group">
131.             <input type="text" id="userInput" placeholder="e.g., racecar, 12321, or
A man, a plan..." />
132.             <button onclick="checkPalindrome()">Check</button>
133.         </div>
134.
135.         <div id="result"></div>
136.     </div>
137.
138.     <script>
139.         function checkPalindrome() {
140.             const inputElement = document.getElementById("userInput");
141.             const resultElement = document.getElementById("result");
142.             const rawText = inputElement.value;
143.
144.             if (!rawText.trim()) {
```



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```
145.         resultElement.className = "error";
146.         resultElement.textContent = "Please enter something to check!";
147.         return;
148.     }
149.
150.     // Clean the string: remove non-alphanumeric characters and convert to
    lowercase
151.     const cleaned = rawText.toLowerCase().replace(/[^a-z0-9]/g, "");
152.
153.     // Reverse the cleaned string
154.     const reversed = cleaned.split("").reverse().join("");
155.
156.     // Compare cleaned string with its reversed counterpart
157.     if (cleaned === reversed) {
158.         resultElement.className = "success";
159.         resultElement.innerHTML = `<strong>${rawText}</strong> is a
    palindrome!`;
160.     } else {
161.         resultElement.className = "error";
162.         resultElement.innerHTML = `<strong>${rawText}</strong> is NOT a
    palindrome.`;
163.     }
164. }
165.
166. // Allow pressing "Enter" key to trigger the check
167. document.getElementById("userInput").addEventListener("keypress",
    function(event) {
168.         if (event.key === "Enter") {
169.             checkPalindrome();
170.         }
171.     });
172. </script>
173.
174. </body>
175. </html>
```

4.Output;



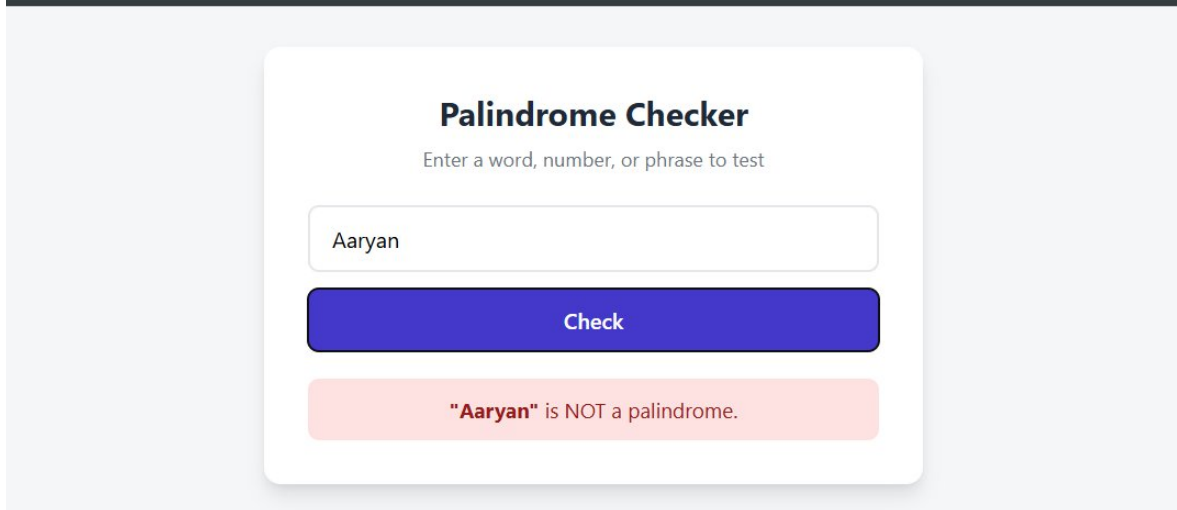
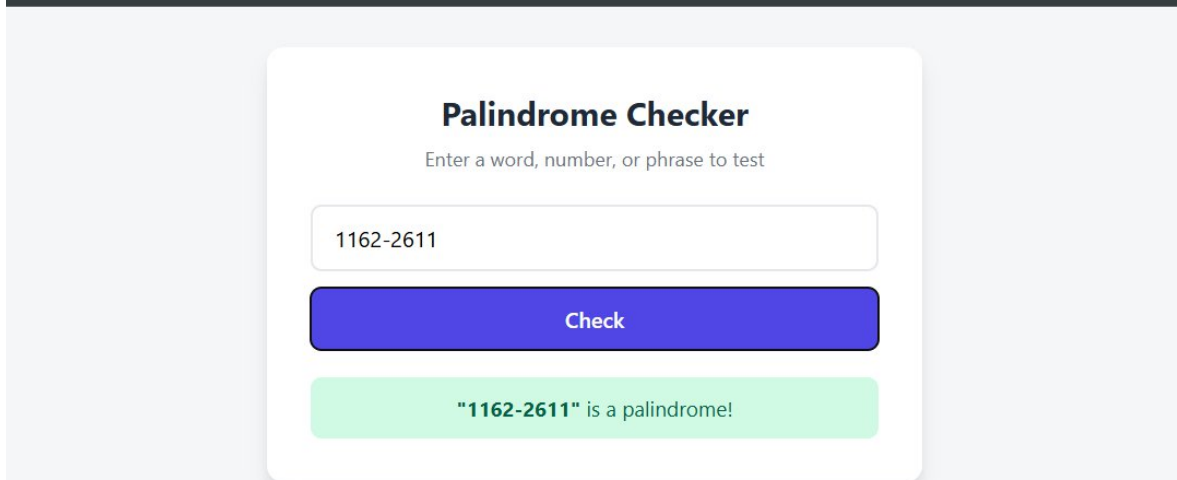
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Case Study 4



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1. Case Study Title: Desing a webpage to check whether the password is Palindrome or not if Palindrome display msg Weak Pass if not display Strong Pass.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-scale=1.0">
8.   <title>Password Palindrome & Strength Checker</title>
9.   <style>
10.    :root {
11.      --bg-color: #0f172a;
12.      --card-bg: #1e293b;
13.      --primary: #6366f1;
14.      --primary-hover: #4f46e5;
15.      --text-main: #f8fafc;
16.      --text-muted: #94a3b8;
17.      --warning-bg: #451a03;
18.      --warning-text: #fde047;
19.      --warning-border: #854d0e;
20.      --success-bg: #064e3b;
21.      --success-text: #6ee7b7;
22.      --success-border: #047857;
23.      --error-bg: #450a0a;
24.      --error-text: #fca5a5;
25.      --error-border: #991b1b;
26.    }
27.
28.    * {
29.      box-sizing: border-box;
30.      margin: 0;
31.      padding: 0;
32.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica,
        Arial, sans-serif;
33.    }
34.
35.    body {
36.      background-color: var(--bg-color);
37.      color: var(--text-main);
38.      display: flex;
39.      justify-content: center;
40.      align-items: center;
41.      min-height: 100vh;
42.      padding: 20px;
43.    }
```



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```
44.  
45. .container {  
46.   background-color: var(--card-bg);  
47.   padding: 32px;  
48.   border-radius: 12px;  
49.   box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.5);  
50.   width: 100%;  
51.   max-width: 480px;  
52.   border: 1px solid #334155;  
53. }  
54.  
55. h1 {  
56.   font-size: 1.5rem;  
57.   margin-bottom: 8px;  
58.   color: var(--text-main);  
59.   text-align: center;  
60. }  
61.  
62. p.subtitle {  
63.   font-size: 0.9rem;  
64.   color: var(--text-muted);  
65.   text-align: center;  
66.   margin-bottom: 24px;  
67. }  
68.  
69. .input-group {  
70.   display: flex;  
71.   flex-direction: column;  
72.   gap: 12px;  
73. }  
74.  
75. .password-wrapper {  
76.   position: relative;  
77.   width: 100%;  
78.   display: flex;  
79.   align-items: center;  
80. }  
81.  
82. input[type="text"] {  
83.   width: 100%;  
84.   padding: 12px 60px 12px 16px;  
85.   font-size: 1.1rem;  
86.   letter-spacing: 1px;  
87.   background-color: #0f172a;  
88.   color: var(--text-main);  
89.   border: 2px solid #334155;  
90.   border-radius: 8px;  
91.   outline: none;
```




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```
92.   transition: border-color 0.2s;
93. }
94.
95.   input:focus {
96.     border-color: var(--primary);
97.   }
98.
99.   .toggle-btn {
100.     position: absolute;
101.     right: 12px;
102.     background: none;
103.     border: none;
104.     color: var(--text-muted);
105.     cursor: pointer;
106.     font-size: 0.85rem;
107.     padding: 4px 8px;
108.     border-radius: 4px;
109.     width: auto;
110.   }
111.
112.   .toggle-btn:hover {
113.     color: var(--text-main);
114.     background-color: transparent;
115.   }
116.
117.   button.btn-submit {
118.     width: 100%;
119.     padding: 12px;
120.     font-size: 1rem;
121.     font-weight: 600;
122.     color: #ffffff;
123.     background-color: var(--primary);
124.     border: none;
125.     border-radius: 8px;
126.     cursor: pointer;
127.     transition: background-color 0.2s;
128.   }
129.
130.   button.btn-submit:hover {
131.     background-color: var(--primary-hover);
132.   }
133.
134.   #result {
135.     margin-top: 20px;
136.     padding: 14px 16px;
137.     border-radius: 8px;
138.     font-size: 0.95rem;
139.     text-align: center;
```



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```
140.         display: none;
141.         word-break: break-word;
142.         border: 1px solid transparent;
143.     }
144.
145.     #result.warning {
146.         display: block;
147.         background-color: var(--warning-bg);
148.         color: var(--warning-text);
149.         border-color: var(--warning-border);
150.     }
151.
152.     #result.success {
153.         display: block;
154.         background-color: var(--success-bg);
155.         color: var(--success-text);
156.         border-color: var(--success-border);
157.     }
158.
159.     #result.error {
160.         display: block;
161.         background-color: var(--error-bg);
162.         color: var(--error-text);
163.         border-color: var(--error-border);
164.     }
165. </style>
166. </head>
167. <body>
168.
169.     <div class="container">
170.         <h1>Password Security Checker</h1>
171.         <p class="subtitle">Check if your password is a vulnerable
172.         palindrome</p>
173.
174.         <div class="input-group">
175.             <div class="password-wrapper">
176.                 <input type="text" id="passwordInput" placeholder="Enter
177.                 password..." />
178.                 <button type="button" class="toggle-btn" id="toggleBtn"
179.                 onclick="toggleVisibility()">Hide</button>
180.             </div>
181.             <button class="btn-submit" onclick="checkPassword()">Check
182.             Password</button>
183.         </div>
184.
185.         <div id="result"></div>
186.     </div>
```



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```
184. <script>
185.     let actualPassword = "";
186.     let isHidden = false;
187.
188.     const inputElement = document.getElementById("passwordInput");
189.
190.     // Intercept input changes to support custom asterisk masking
191.     inputElement.addEventListener("input", function(e) {
192.         if (isHidden) {
193.             // If characters were added or deleted
194.             const diff = e.target.value.length - actualPassword.length;
195.
196.             if (diff > 0) {
197.                 // Characters added
198.                 const addedChars = e.target.value.slice(actualPassword.length);
199.                 actualPassword += addedChars;
200.             } else if (diff < 0) {
201.                 // Characters removed (Backspace)
202.                 actualPassword = actualPassword.slice(0, e.target.value.length);
203.             }
204.
205.             // Render asterisks in the input display
206.             inputElement.value = "*".repeat(actualPassword.length);
207.         } else {
208.             actualPassword = e.target.value;
209.         }
210.     });
211.
212.     function toggleVisibility() {
213.         const btn = document.getElementById("toggleBtn");
214.
215.         if (!isHidden) {
216.             // Switch to Hidden mode -> replace characters with *
217.             isHidden = true;
218.             inputElement.value = "*".repeat(actualPassword.length);
219.             btn.textContent = "Show";
220.         } else {
221.             // Switch to Visible mode -> show raw password
222.             isHidden = false;
223.             inputElement.value = actualPassword;
224.             btn.textContent = "Hide";
225.         }
226.     }
227.
228.     function checkPassword() {
229.         const resultElement = document.getElementById("result");
230.         const password = actualPassword;
231.
```



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```
232.         if (!password) {
233.             resultElement.className = "error";
234.             resultElement.textContent = "Please enter a password first!";
235.             return;
236.         }
237.
238.         // Reverse exact password
239.         const reversedPassword = password.split("").reverse().join("");
240.
241.         // Check if exact palindrome
242.         if (password === reversedPassword) {
243.             resultElement.className = "warning";
244.             resultElement.innerHTML = "⚠ <strong>Weak password!</strong> It is
a palindrome. Choose a strong password.";
245.         } else {
246.             resultElement.className = "success";
247.             resultElement.innerHTML = "👍 <strong>It is a strong non-palindrome
password.</strong>";
248.         }
249.     }
250.
251.     // Trigger check on Enter key press
252.     inputElement.addEventListener("keypress", function(event) {
253.         if (event.key === "Enter") {
254.             checkPassword();
255.         }
256.     });
257. </script>
258.
259. </body>
260. </html>
```

3.Output:



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Browser: Password Palindrome & Strengi x + | Ask Gemini | - □ ×

Address Bar: 127.0.0.1:5500/Exp4/exp4_Aaryan_Kawale_1162CS.html ☆ 🗑️ 👤 ⋮

Password Security Checker

Check if your password is a vulnerable palindrome

Hide

Check Password

🔒 It is a strong non-palindrome password.

Browser: Password Palindrome & Strengi x + | Ask Gemini | - □ ×

Address Bar: 127.0.0.1:5500/Exp4/exp4_Aaryan_Kawale_1162CS.html ☆ 🗑️ 👤 ⋮

Password Security Checker

Check if your password is a vulnerable palindrome

Hide

Check Password

⚠️ Weak password! It is a palindrome. Choose a strong password.